

Each question is worth 5 points for a total of 115 points. The extra credit can get you an additional 5 points making the highest possible score 120 out of 115. Write answers CLEARLY. If I cannot determine your answer, the problem will receive a score of 0.

Problem 1

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $\cos(2\theta) = \cos \theta$.

Problem 2

To measure the height of Lincoln's caricature on Mt. Rushmore, two sightings 400 feet from the base of the mountain are taken. If the angle of elevation to the bottom of Lincoln's face is 30° and the angle of elevation to the top is 45° , what is the height of Lincoln's face?

Problem 3

Solve the triangle $a = 6$, $b = 12$, $\alpha = 30^\circ$.

Problem 4

Find the area of the triangle $a = 4$, $b = 12$, $\gamma = \frac{\pi}{3}$.

Problem 5

Find the area of the triangle $a = 2$, $b = 2$, $c = 2$.

Problem 6

The displacement d (in meters) of an object at time t (in seconds) is given by the equation $d = 6 \cos(\pi t)$. Find the amplitude, period, and frequency.

Problem 7

Find the exact value $\frac{\tan 20^\circ + \tan 25^\circ}{1 - \tan 20^\circ \tan 25^\circ}$.

Problem 8

Find the exact value $\sin[\sin^{-1} \frac{3}{5} - \cos^{-1}(-\frac{4}{5})]$.

Problem 9

Find the exact value $\tan \frac{9\pi}{8}$.

Problem 10

Establish the identity and SHOW ALL YOUR WORK.

$$\tan(3\theta) = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}.$$

Problem 11

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \sin \theta + 3 = 2$.

Problem 12

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \sin^2 \theta - \sin \theta - 1 = 0$.

Problem 13

Find the exact value of $\tan(100^\circ)\cot(100^\circ)$.

Problem 14

Determine the amplitude and period of $y = -5\sin(\pi x)$.

Problem 15

Graph $y = 2\csc(2x - \frac{\pi}{2})$.

Problem 16

Find the exact value of $\csc(\cos^{-1} \frac{3}{5})$.

Problem 17

Establish the identity and SHOW ALL YOUR WORK.

$$\frac{1 - \sin \theta}{1 + \sin \theta} = (\sec \theta - \tan \theta)^2.$$

Problem 18

Show that $\sec(\tan^{-1} v) = \sqrt{1 + v^2}$.

Problem 19

Convert each angle in degrees to radians. Express your answer as a multiple of π .

- a) 30°
- b) 90°
- c) 315°
- d) 125°

Problem 20

Let s denote the length of the arc of a circle of radius r subtended by the central angle θ . Find the missing quantity.

a) $r = 17$ meters, $\theta = 3$, $s = ?$

b) $r = 2$ inches, $\theta = ?$, $s = \pi$ inches

c) $r = ?$, $\theta = \frac{\pi}{3}$, $s = 2\pi$ feet

Problem 21

Find the exact value.

a) $\sin \frac{13\pi}{4}$

b) $\cos \frac{7\pi}{6}$

c) $\tan \frac{9\pi}{4}$

Problem 22

The point $(-5, 12)$ is on the terminal side of an angle θ . Find the exact value of the six trigonometric functions.

Problem 23

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \cos^2 \theta + \cos \theta - 1 = 0$.

Problem 24

Let $\sin \theta = \frac{3}{4}$ where θ is in the first quadrant. Find the exact value of the other five trigonometric functions.

Extra Credit

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $\sqrt{3} \sin \theta + \cos \theta = 1$.